Measure Theoretic Probability
Lesson 13
Tail σ-Fields and Kolmogorov's Zero–One Law

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Outline

1. The Tail σ-Field
2. Examples of Tail Events
3. Kolmogorov's Zero–One Law
4. Proof of the Zero–One Law
5. Summary and Exercises

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Setting and Definition

Setting

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $A_1, A_2, \cdots \in \mathcal{F}$. We want a σ-field describing what the sequence does **"at the end"**, no matter where we start looking.

Definition (tail σ-field)

$$\mathcal{F}_T = \bigcap_{n=1}^{\infty} \sigma(A_n, A_{n+1}, A_{n+2}, \dots).$$

Events in $\mathcal{F}_T$ are called **tail events**.

Remark

Each $\sigma(A_n, A_{n+1}, \dots)$ is a σ-field, and an intersection of σ-fields is a σ-field, so $\mathcal{F}_T$ is indeed a σ-field. Also $\mathcal{F}_T \subseteq \mathcal{F}$.

What Does a Tail Event Look Like?

Definition

Let $A \in \mathcal{F}_T$. Then A lies in $\sigma(A_n, A_{n+1}, \dots)$ for **every** n:

- $n = 1$: A is built from $A_1, A_2, A_3, \dots$
- $n = 2$: A is built from $A_2, A_3, A_4, \dots$ (we did not need A_1)
- any N: A is built from $A_N, A_{N+1}, \dots$

"Built from" means using complements, countable unions and countable intersections.

$$\begin{array}{c} \sigma(A_1, A_2, \dots) \\ \supseteq \\ \sigma(A_2, A_3, \dots) \\ \supseteq \\ \sigma(A_3, A_4, \dots) \\ \supseteq \\ \vdots \\ \downarrow \\ \mathcal{F}_T \end{array}$$

Intuition: limits of real numbers

Dropping the first N terms of a real sequence does not change its limit. Likewise, a tail event is unaffected by changing or discarding any **finite** number of the A_n (not only the first ones: eventually the index N passes the largest subscript we want to remove).

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Example 1: Increasing Sequences

Claim

If $A_n \uparrow A$ (that is, $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$), then $A \in \mathcal{F}_T$.

Proof

Because the sets are nested, discarding the first $m-1$ sets does not change the union: for **every** $m \geq 1$,

$$A = \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=m}^{\infty} A_n.$$

So A is built from $A_m, A_{m+1}, \ldots$, i.e. $A \in \sigma(A_m, A_{m+1}, \ldots)$. This holds for every $m \geq 1$, hence

$$A \in \bigcap_{m \geq 1} \sigma(A_m, A_{m+1}, \ldots) = \mathcal{F}_T. \qquad \square$$

Example 2: Decreasing sequences (exercise)

If $A_n \downarrow A$ (that is, $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_n A_n$), then $A \in \mathcal{F}_T$. The proof is the same, with intersections in place of unions.

Example 3: lim sup and Example 4: lim inf

Claim

For **any** sequence $A_1, A_2, \cdots \in \mathcal{F}$ (no monotonicity), $\limsup_n A_n \in \mathcal{F}_T$.

Proof

Put $B_N = \bigcup_{m \geq N} A_m$. Then $B_1 \supseteq B_2 \supseteq \cdots$ is a **decreasing** sequence, and

$$\limsup_n A_n = \bigcap_{N=1}^{\infty} B_N.$$

Fix n. Dropping the first $n-1$ sets of a decreasing sequence does not change its intersection, so

$$\limsup_m A_m = \bigcap_{N \geq n} \bigcup_{m \geq N} A_m \in \sigma(A_n, A_{n+1}, \dots),$$

since every A_m appearing here has $m \geq N \geq n$. As n was arbitrary, $\limsup_n A_n \in \mathcal{F}_T$. □

Example 4: $\liminf_n A_n \in \mathcal{F}_T$ **(exercise)**

Write $\liminf_n A_n = \bigcup_N \bigcap_{m \geq N} A_m$. The inner sets **increase** with N; argue exactly as above.

Consequence

$\{A_n \text{ i.o.}\}$ and $\{A_n \text{ f.o.}\}$ are tail events. This is what links the Borel–Cantelli lemmas to this lesson.

A Non-Example (and a Preview)

Is A_1 a tail event?

In general, **no**: A_1 is built from A_1, but it need not be built from $A_2, A_3, \ldots$. Whether the first round was a success says nothing about the "end" of the sequence.

Sanity check from the next theorem

If the A_n are independent with $\mathbb{P}(A_1) = \frac{1}{2}$, then A_1 **cannot** be a tail event:
Kolmogorov's law (We will look at that) forces every tail event to have probability 0 or 1.

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The Theorem

Kolmogorov's 0–1 Law

Let $A_1, A_2, \cdots \in \mathcal{F}$ be **independent** events and let $\mathcal{F}_T$ be their tail σ-field. Then for every $A \in \mathcal{F}_T$,

$$\mathbb{P}(A) \in \{0, 1\}.$$

What the law does and does not say

- Every tail event is almost sure or almost impossible.
- It does **not** tell you which of 0 or 1 occurs; that is usually hard.
- By Example 3, $\mathbb{P}(A_n \text{ i.o.}) \in \{0, 1\}$ for independent events. Borel–Cantelli I and II **decide** which one, when $\sum \mathbb{P}(A_n)$ converges or diverges.

Why Independence Cannot Be Dropped

A counterexample

Take $A \in \mathcal{F}$ with $0 < \mathbb{P}(A) = p < 1$ and set $A_n = A$ for every n.

Computation

- $\sum_n \mathbb{P}(A_n) = \sum_n p = \infty$ (we add a constant $p > 0$ forever).
- $\limsup_n A_n = \bigcap_n \bigcup_{m \geq n} A = A$, so by Example 3 it is a tail event.
- But $\mathbb{P}(\limsup_n A_n) = p \in (0, 1)$.

Conclusion

Without independence, a lim sup (a tail event) can have probability strictly between 0 and 1. Also, Borel–Cantelli II is not just "a special case" of Kolmogorov: the zero–one law gives 0 or 1 but does not say which, whereas BC II gives the answer 1.

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Recalling from earlier Lessons

Grouping lemma

If $A_1, A_2, \dots$ are independent, then A_1 is independent of $\sigma(A_2, A_3, \dots)$. Applying it after discarding the indices $2, \dots, n-1$, A_1 is independent of $\sigma(A_n, A_{n+1}, \dots)$.

Extension

More generally, for every $n \geq 2$ the finite family $A_1, \dots, A_{n-1}$ is independent of the σ-field

$$\sigma(A_n, A_{n+1}, \dots),$$

meaning each event of $\sigma(A_1, \dots, A_{n-1})$ is independent of each event of $\sigma(A_n, A_{n+1}, \dots)$.

Main theorem

If $\mathcal{P}$ is a π-system and B is independent of every member of $\mathcal{P}$, then B is independent of $\sigma(\mathcal{P})$.

Proof of Kolmogorov's Law (1/2): Independence from Everything

Step 1: A is independent of each initial block

Let $A \in \mathcal{F}_T$ and fix $n \geq 2$. Since $A \in \sigma(A_n, A_{n+1}, \dots)$, the Extension says

$$A \text{ is independent of } \sigma(A_1, \dots, A_{n-1}).$$

In particular, A is independent of every finite intersection $A_{i_1} \cap \cdots \cap A_{i_k}$ (choose $n > i_k$).

Step 2: A is independent of the whole sequence

Let

$$\mathcal{P} = \{A_{i_1} \cap \cdots \cap A_{i_k} : \ k \geq 1, \ i_1 < \cdots < i_k\}.$$

This is a π-system (the intersection of two members is again a finite intersection) and $\sigma(\mathcal{P}) = \sigma(A_1, A_2, \dots)$. By Step 1, A is independent of every member of $\mathcal{P}$. By the main theorem of Lesson 10,

$$A \text{ is independent of } \sigma(A_1, A_2, A_3, \dots).$$

Proof of Kolmogorov's Law (2/2): A Is Independent of Itself

Step 3: the punchline

But $A \in \mathcal{F}_T \subseteq \sigma(A_1, A_2, \dots)$ (take $n = 1$ in the definition of $\mathcal{F}_T$). So A is an event of the σ-field from which it is independent. Taking the pair (A, A):

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)\,\mathbb{P}(A) = \mathbb{P}(A)^2.$$

Step 4: solve

$x = x^2$ means $x(1 - x) = 0$, so $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$. □

The idea in one line

A tail event is independent of every finite part of the sequence, hence of all of it; but it is itself determined by the sequence, so it is independent of itself.

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Summary

Key results

Concept / Result	Content
Tail σ-field	$\mathcal{F}_T = \bigcap_n \sigma(A_n, A_{n+1}, \dots)$; events in $\mathcal{F}_T$ are tail events
Meaning	a tail event can be built from $A_N, A_{N+1}, \dots$ for every N
Examples	$A_n \uparrow A$, $A_n \downarrow A$, $\limsup_n A_n$, $\liminf_n A_n$ are all in $\mathcal{F}_T$
Kolmogorov 0–1 law	A_n independent, $A \in \mathcal{F}_T \Rightarrow \mathbb{P}(A) \in \{0, 1\}$
Proof idea	A independent of $\sigma(A_1, A_2, \dots) \ni A$, so $\mathbb{P}(A) = \mathbb{P}(A)^2$
Warnings	the law does not say which value; independence is essential

Exercises

1. Prove Example 2: if $A_n \downarrow A$, then $A \in \mathcal{F}_T$.
2. Prove Example 4: $\liminf_n A_n \in \mathcal{F}_T$ for any sequence in $\mathcal{F}$.
3. Let A_n be independent with $\mathbb{P}(A_n) = 1/n^2$. What are the possible values of $\mathbb{P}(A_n \text{ i.o.})$ by Kolmogorov, and which one does Borel–Cantelli give? Repeat for $\mathbb{P}(A_n) = 1/n$.
4. Show that if A_n are independent and $\mathbb{P}(A_1) = \frac{1}{2}$, then A_1 is not a tail event. (Hint: use the 0–1 law.)
5. Let $A \in \mathcal{F}$ with $0 < \mathbb{P}(A) < 1$ and $A_n = A$ for all n. Describe $\mathcal{F}_T$ and verify that Kolmogorov's conclusion fails. Which hypothesis is violated?
6. Explain why changing finitely many of the events A_n does not change $\mathcal{F}_T$.

Outlook

Next lesson

Random variables. We move from events to functions on Ω, and the same tail construction applies to sequences of random variables.

Why this matters

The strongest applications of Kolmogorov's 0–1 law are in deciding **convergence** of sequences and series of random variables: the event that such a sequence converges is a tail event, so for independent terms it has probability 0 or 1.